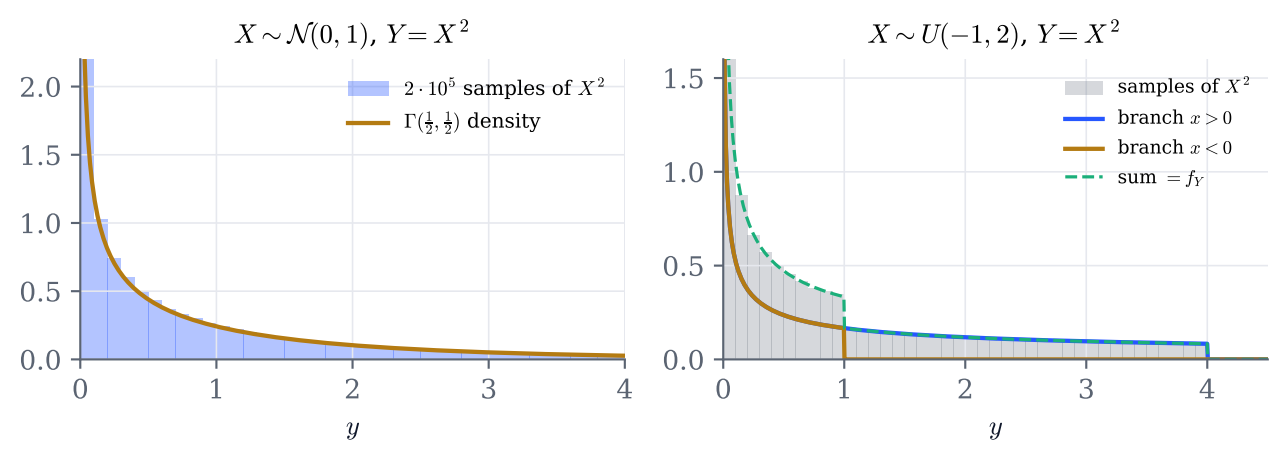
$X \sim \mathcal{N}(0,1)$, $Y = X^2$
$2 \cdot 10^5$ samples of X^2
$\Gamma(\frac{1}{2}, \frac{1}{2})$ density
2.0
1.5
1.0
0.5
0.0
0
1
2
3
4
y
$X \sim U(-1,2)$, $Y = X^2$
samples of X^2
branch $x > 0$
branch $x < 0$
sum $= f_Y$
1.5
1.0
0.5
0.0
0
1
2
3
4
y